



Predicting the Decline of US Tourism

Motivation



Recent Travel



The World Cup



Booking Rates



Visa Counts

Related Work

Núñez et al. (2024)

Machine learning applied to tourism: A systematic review. What impact ML is having on tourism.

Outcome

Models based on neural networks account for 15% of the research applied, which suggests that neural networks have the largest impact on the tourism industry.

Liu, Tseng & Tseng (2018)

Big Data analytics for forecasting tourism destination arrivals with the applied Vector Autoregression model based on weather.

Outcome

Shockingly weather was found to have no correlation with actual arrivals of the studied destination.



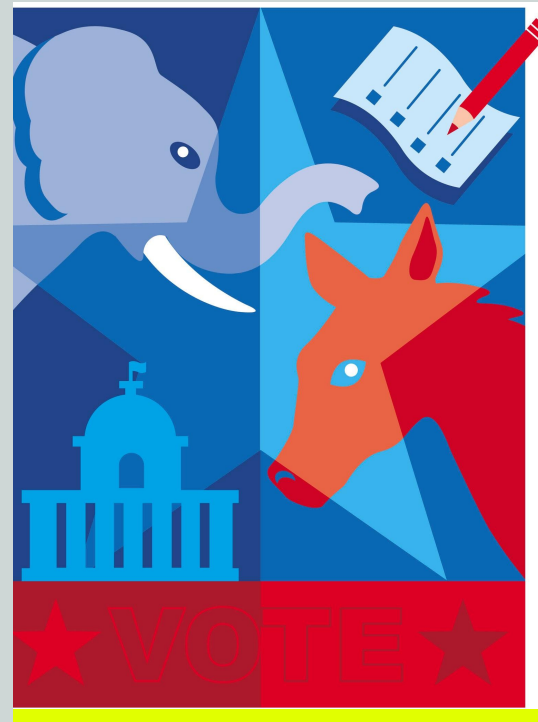
Our Data



I-94 arrivals program



Tourism and GDP
Statistics



Historic US Political
Party Control



Armed Conflicts &
Political Unrest

I-94 Arrivals Program



What is the data

- The National Travel and Tourism Office's Arrivals



Why did we choose it

- To best select a range of data to focus on and baseline



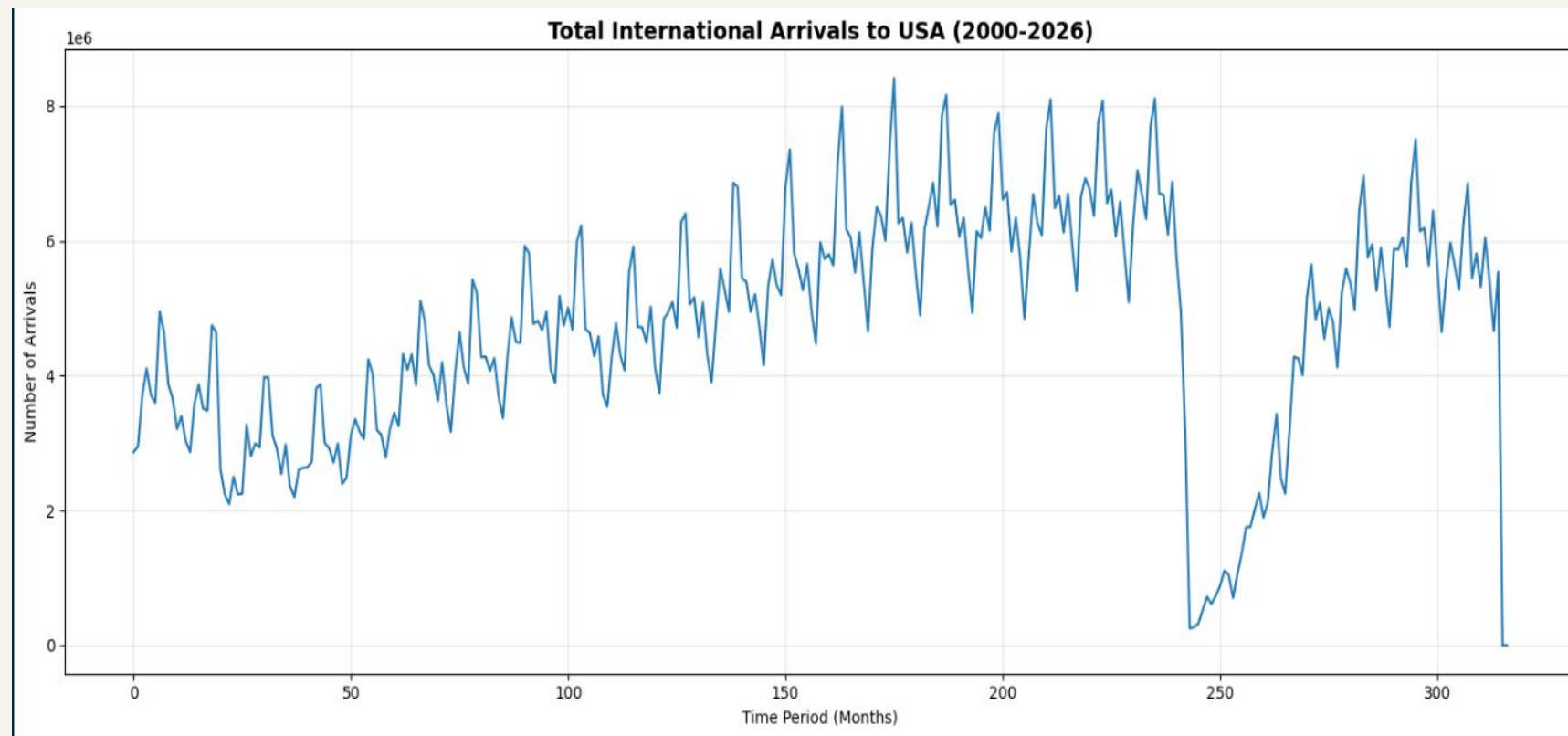
Difficulties

- Null management and COVID outlier years



Key Features

- Arrival count by time period



Tourism and GDP Database



What is the data

- Tourism data gathered and approved by the UN.



Why did we choose it

- There were details like GDP and GDP related to tourism



Difficulties

- The large data sets caused load issues



Key Features

- Years, counts, financial details and financial totals



Historic US Political Party Control



What is the data

- US House of Representatives Party Divisions History



Why did we choose it

- Political indicators to add nuance to the drivers



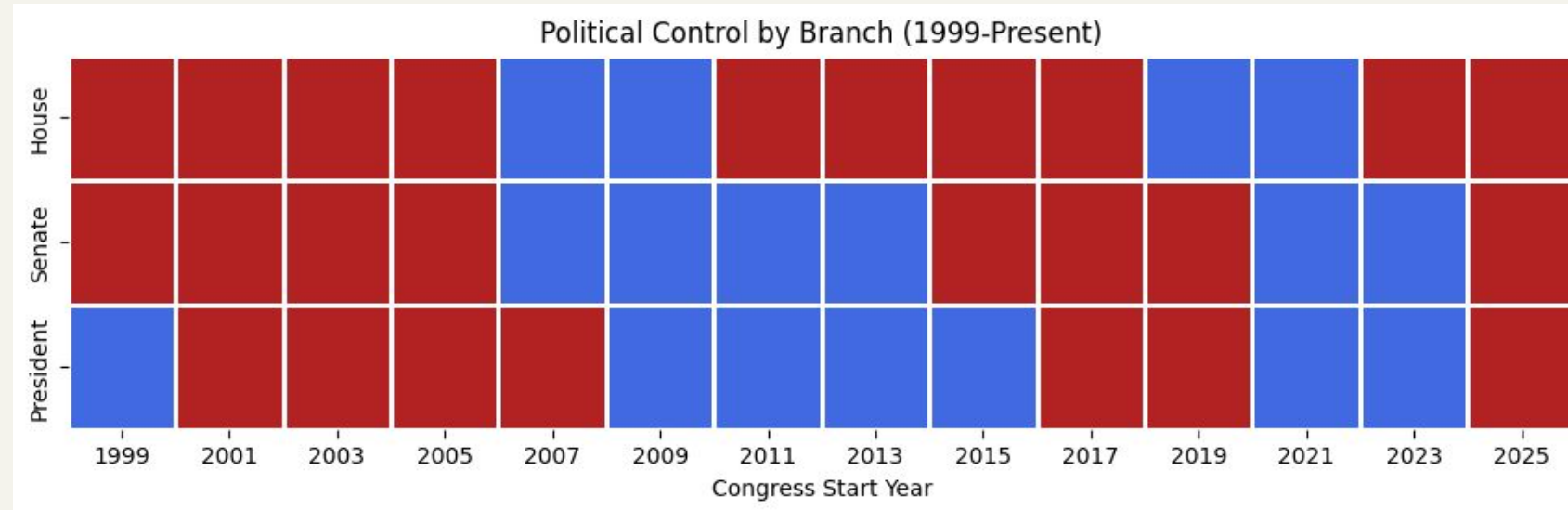
Difficulties

- The insight we were trying to grasp wouldn't be possible on this feature alone



Key Features

- Political trend over the years



Armed Conflicts & Political Unrest



What is the data

- Impartial violent conflict and protest in all countries.



Difficulties

- Data only includes events from 2020-2025



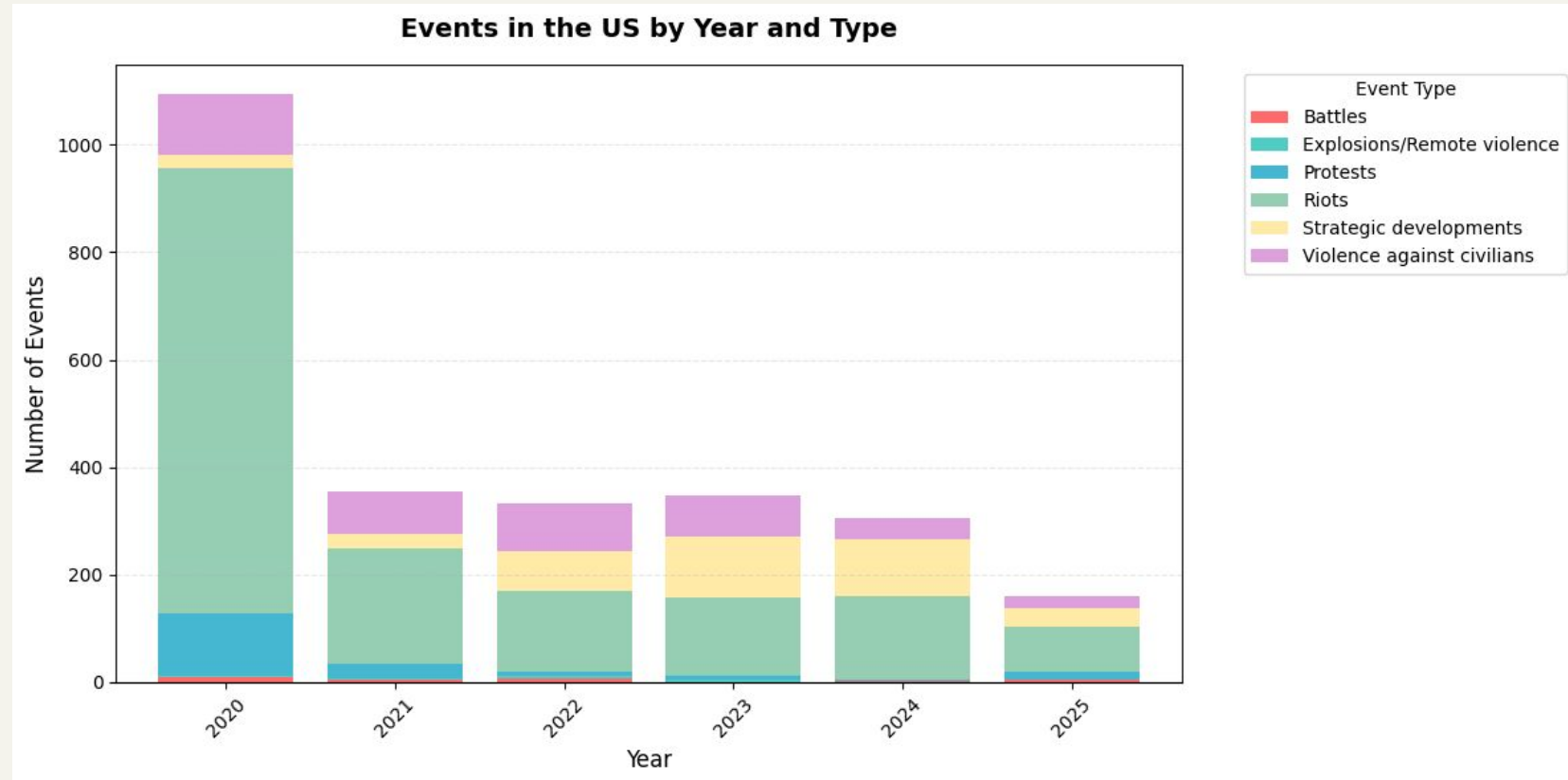
Why did we choose it

- To see how specific events would impact our model



Key Features

- Event types, severity and impact

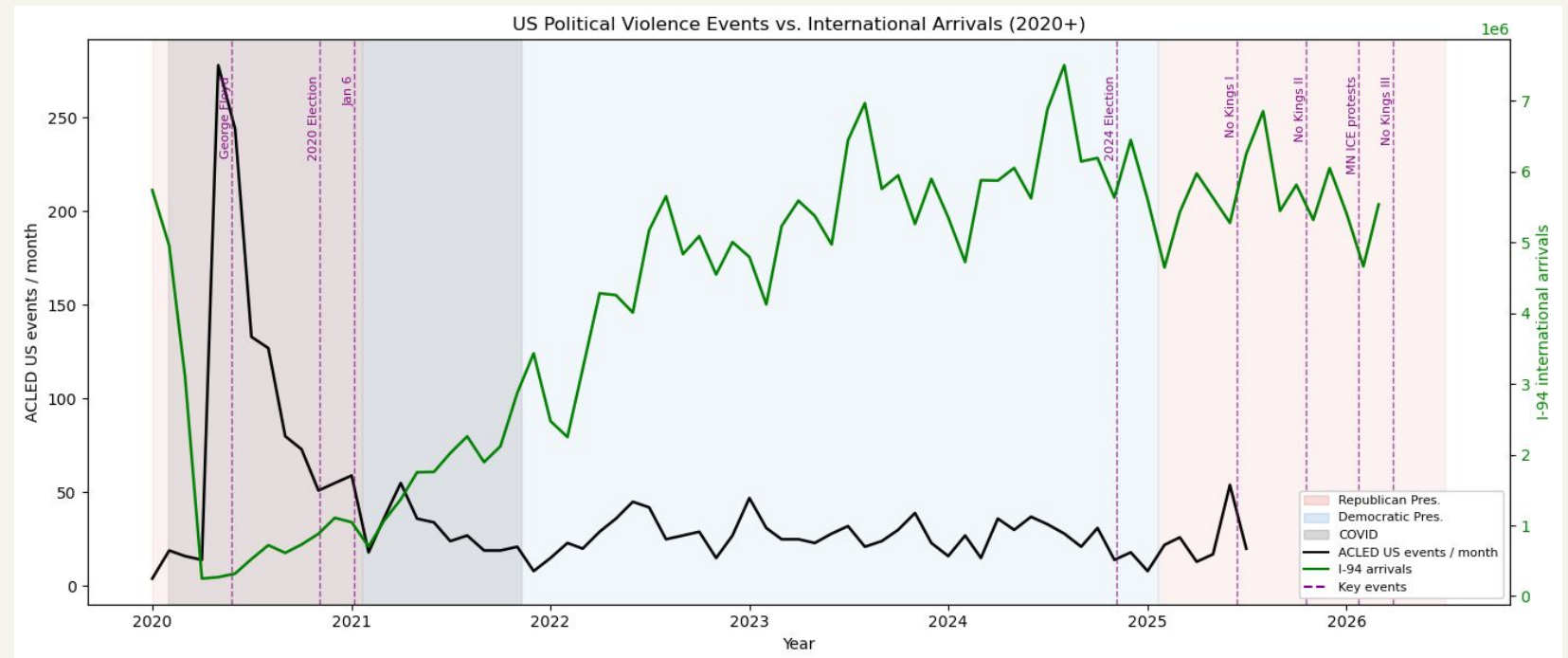


The Experiment

ACLELED Relationship w/ I-94 Arrivals

First run through

feature	coefficient
is_rep_house	0.000000
is_rep_senate	0.000000
is_rep_president	0.000000
is_unified	0.000000
house_changed	0.000000
senate_changed	0.000000
pres_changed	0.000000
is_election_year	0.000000
tourism_value	0.000000
tourism_pct_change	0.000000
gdp_value	0.000000
gdp_pct_change	0.000000
acled_incidents	-440099.833248



Feature Engineering

Initial Features

Congressional

is_rep_house, is_rep_senate,
is_rep_presidentis_unified, house_changed,
senate_changed, pres_changed
ls_election_year

Tourism

tourism_value, tourism_pct_change, gdp_value,
gdp_pct_change

Politically-Violent Events

acled_incidents, acled_incidents,
acled_fatal_events, acled_total_fatalities,
acled_any_event, acled_any_fatal_event,
acled_high_activity_flag, acled_events_lag_1,
acled_fatal_events_lag_1, acled_events_lag_2,
acled_fatal_events_lag_2, acled_events_roll_3,
acled_fatalities_roll_3, acled_states,
acled_event_types

Intermediate Features

Congressional

is_rep_house, is_rep_senate,
is_rep_president, is_unified,
house_changed, senate_changed,
pres_changed
ls_election_year

Tourism

tourism_value, tourism_pct_change,
gdp_value, gdp_pct_change

Politically-Violent Events

acled_incidents, acled_events_roll_3,
acled_events_lag_1

Time-Based

arrivals_lag_1, arrivals_lag_2,
arrivals_lag_12, arrivals_roll_3,
month_sin, month_cos

Final Features!

- arrivals_lag_1
- arrivals_lag_12
- arrivals_roll_3
- month_sin
- month_cos
- acled_incidents
- acled_events_roll_3
- acled_events_lag_1

Experiment Design

Why not use RNN?

We decided not to use a neural network (RNN) due to the limited number of labels available. Any deep-learning style approach will most likely result in overfitting on the training data due to the limited sample size of the I-94 arrival data (315 total).

Hypothesis

Decision Tree based models using either Boosting or Random Forests will generate the most accurate results

Experiment Overview

Mean-value prediction and linear regression will establish our baseline models. Careful feature selection followed by hyperparameter tuning on our Random Forest/XGBoosting models will maximize accuracy.

Design Considerations

Constraints:

- COVID-19 pandemic structural break
- Datasets account for different timeframes
 - Any absences in data could not be replaced with NaN or zeros
- Time-series nature of the data prevents randomized training/test split

Decisions:

- Use features and model selection to contextualize the pandemic
- Modeling was done on data from 2009-2026 to maximize feature inclusion, and from 2020–2025 including ACLED
- Training/test split was defined by a “split-date” rather than randomization to avoid leakage (split chosen for January 1st, 2022).

Baseline Models

Mean-value Predictor (Performance Baseline)

- Predicts the mean (5,286,071 arrivals) from the training data every time
- Results:
 - RMSE: 1,125,259
 - MAE: 841,561

Initial Linear Regression (Feature Coefficients)

arrivals_roll_3	1.295685e+06
arrivals_lag_1	5.275817e+05
arrivals_lag_12	5.112898e+05
tourism_value	4.575113e+05
gdp_pct_change	3.026004e+05
is_unified	6.017291e+04
pres_changed	6.017291e+04
house_changed	3.859956e+04
gdp_value	2.021625e+04
is_rep_house	1.447328e+04
is_election_year	-1.324918e+04
senate_changed	-1.727868e+04
month_sin	-2.696411e+04
tourism_pct_change	-3.405852e+04
is_rep_senate	-1.021430e+05
month_cos	-1.995782e+05
is_rep_president	-2.139809e+05
arrivals_lag_2	-7.128412e+05

Why Decision Trees

Advantages:

- Utilizes time-based events/predictors as features
- Captures nonlinear relationships
- Don't require a large dataset for learning

Considerations:

- Careful feature engineering required for success
- Leakage from training to test dataset if chronological order is not preserved

Random Forest / XGBoost

Random Forest

Parameters:

- `n_estimators=500`,
- `max_depth=3`,
- `min_samples_leaf=2`,
- `max_features=0.8`

Results:

- RMSE : 594,103
- MAE : 460,445
- R^2 : 0.7197

XGBoost

Parameters:

- `n_estimators=250`,
- `max_depth=2`,
- `learning_rate=0.03`,
- `min_child_weight=2`,
- `subsample=0.8`,
- `colsample_bytree=0.9`,
- `reg_lambda=5`

Results:

- RMSE : 557,709
- MAE : 406,561
- R^2 : 0.7530

Tuning / Optimization

Hyperparameter tuning was done using **RandomizedSearchCV** with **TimeSeriesSplit**.

- Chronological order is preserved
- Tuning is only done on training data; test data is avoided to preserve evaluation

Random Forest

Results:

- RMSE: 594,103 -> 780,255
- MAE: 460,445 -> 670,333
- R^2 : 0.7197 -> 0.5166

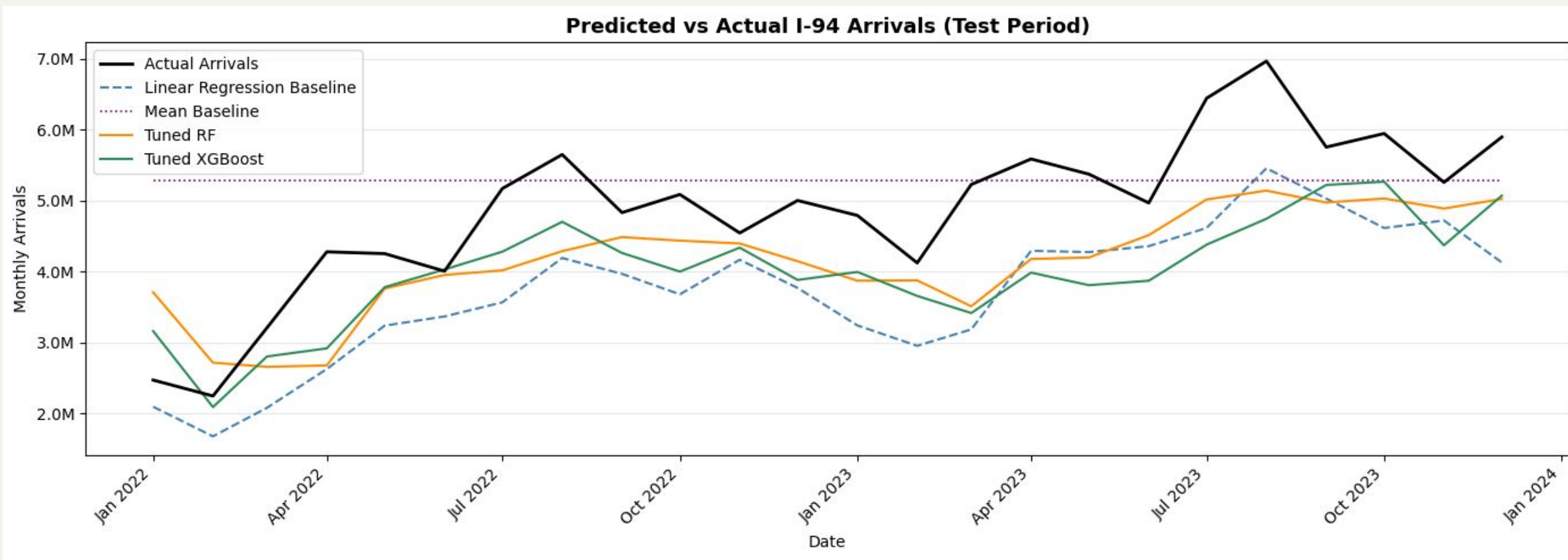
XGBoost

Results:

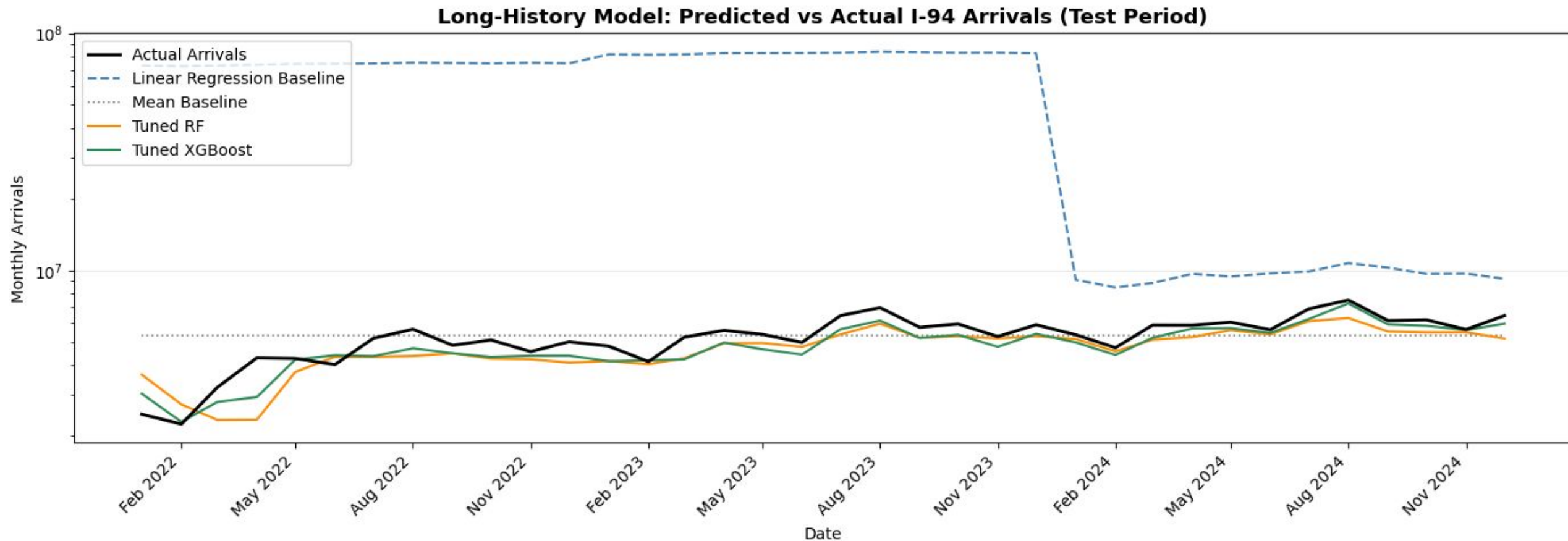
- RMSE : 557,709 -> 581,359
- MAE : 460,561 -> 499,689
- R^2 : 0.7530 -> 0.7316



Initial Test Predictions



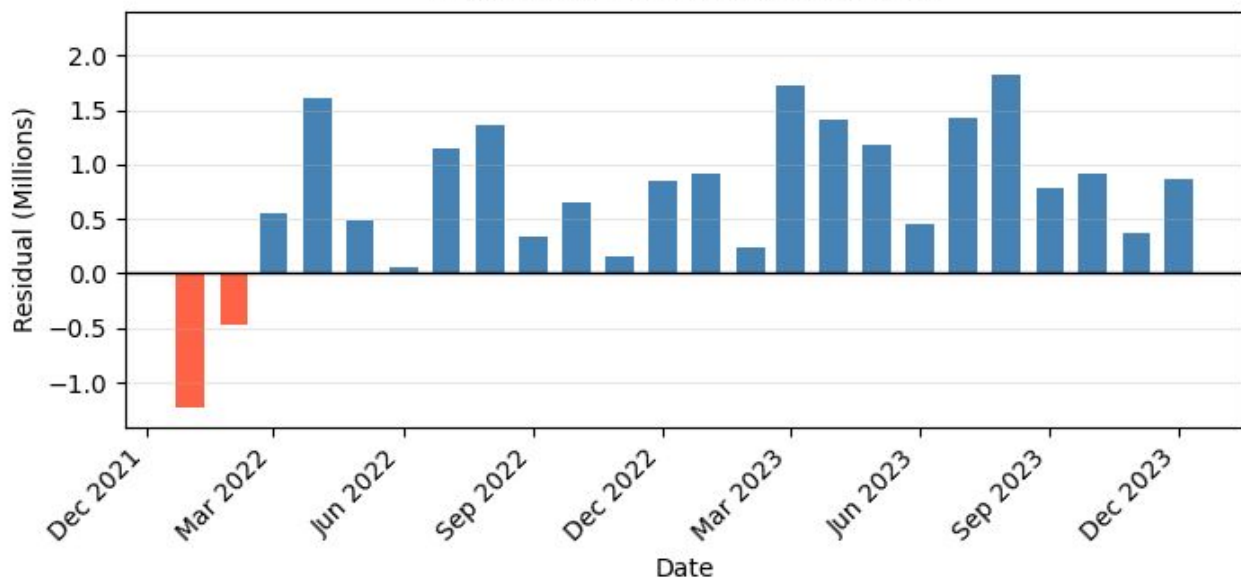
Best Model: XGBoost



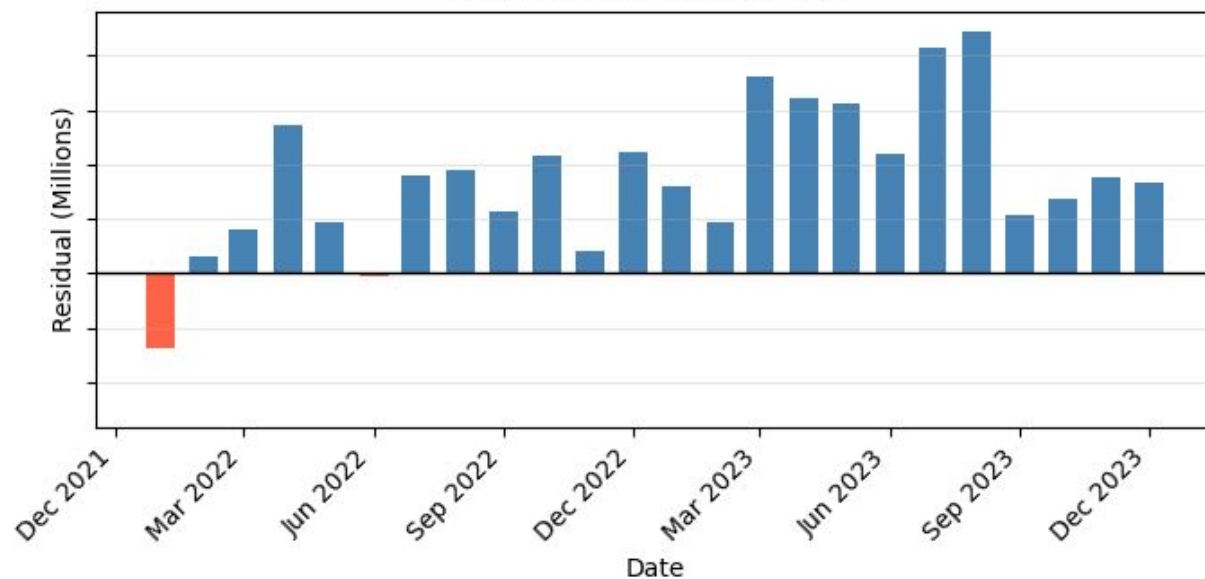
Test Data: Residuals

Prediction Residuals — Actual minus Predicted

Residuals: Tuned Random Forest



Residuals: Tuned XGBoost



Conclusions / Future Work

- Feature engineering resulted in greater performance increases than hyperparameter tuning
 - Introduction of time-based features yielded the best results
 - Pre-tuned Random Forest (**RMSE / MAE / R²**)
 - Initial features: 3,599,063 / 3,534,151 / -23.73
 - Final features: **594,103 / 460,445 / 0.7197**
- Features rarely matched in timeline, resulting in narrower training / test datasets
- The effects of the COVID-19 pandemic were inadequately captured by the model
- Further testing can build on features derived from centralized datasets, reducing risks of confounding variables and timeline gaps

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